

Roll No.

TOE-502

B. TECH. (EE) (FIFTH SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2022

COMPUTER BASED NUMERICAL AND STATISTICAL TECHNIQUES

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) Discuss Inherent Error. Round-off the number 75462 to four significant digits and then calculate the absolute error and percentage error. (CO1)

OR

- (b) If $R = 10 x^3 y^2 z^2$ and errors in, x, y, z are 0.03, 0.01, 0.02 respectively at $x = 3, y = 1, z = 2$. Calculate the absolute error and percentage relative error in evaluating R . (CO1)

2. (a) The negative root of the smallest magnitude of the equation : (CO1)

$$f(x) = 3x^3 + 10x^2 + 10x + 7 = 0$$

is to be obtained.

- (i) Find an interval of unit length which contains this root.
(ii) Perform two iterations of the bisection method.

P. T. O.

OR

(b) Solve by Muller's method :

(CO1)

$$x^3 + 2x^2 + 10x - 20 = 0$$

by taking $x = 0, x = 1, x = 2$ as initial approximations.

3. (a) Apply False-position method to find the smallest positive root of the equation $x - e^{-x} = 0$ correct to three decimal places. Take initial values as 0.56 and 0.58. (CO2)

OR

- (b) Using Newton-Raphson method, find the real root of the equation $3x = \cos x + 1$ correct to four decimal places. Take 0 and 1 as initial value. (CO2)

4. (a) Explain Interpolation. Differentiate between argument and entry. (CO2)

OR

- (b) The population of a town in the decimal census was as given below. Estimate the population for the year 1895 : (CO2)

Year x	Population y (in thousands)
1891	46
1901	66
1911	81
1921	93
1931	101

(3)

5. (a) The population of a town is as follows :

(CO2)

Year	Population (in lakhs)
1921	20
1931	24
1941	29
1951	36
1961	46
1971	51

Estimate the increase in population during the period 1955 to 1961.

OR

(b) Apply a central difference formula to obtain $f(32)$ given that : (CO2)

$$f(25) = 0.2707 \quad f(35) = 0.3386$$

$$f(30) = 0.3037 \quad f(40) = 0.3794.$$